Lecture 14

Thursday, 2 March 2023 10:42 AM

1) $A_2(g) + 2* \rightleftharpoons 2A^*$

2) $B_2(g) + 2* \rightleftharpoons 2B^*$

3) $A^* + B^* \rightleftharpoons AB^* + *$

4) $AB^* + B^* \rightleftharpoons AB_2^* + *$

→ Contd on previous lecture notes.

5) $AB_{(g)} + * \rightleftharpoons AB^*$

6) $AB_2(g) + * \rightleftharpoons AB_2^*$

## Scaling Relations :

1) $N_2(g) + 2* \rightleftharpoons 2N^*$

2) $N^* + H^* \rightleftharpoons NH^* + *$ QE

3) $NH^* + H^* \rightleftharpoons NH_2^* + *$

4) $NH_2^* + H^* \rightleftharpoons NH_{3(g)} + 2*$ QE

5) $H_2(g) + 2* \rightleftharpoons 2H^*$ QE

} These don't happen in any specific sequence

$2r_1 = r_2$ ; $r_2 = r_3$ , $r_3 = r_4$

↳ $2r_1 = r_3$

$$2\left(k_1 P_{N_2} \theta_*^2 - k_{-1} \theta_N^2\right) = k_3 \theta_{NH} \theta_H - k_{-3} \theta_{NH_2} \theta_*$$

$$K_2 = \frac{\theta_{NH}\theta_*}{\theta_N \theta_H} \quad ; \quad K_4 = \frac{\theta_*^2 \dfrac{P_{NH_3}}{p^\circ}}{\theta_{NH_2}} \quad ; \quad K_5 = \frac{\theta_H^2}{\theta_*^2 P_{H_2}/p^\circ}$$

} "full solution"

$$\frac{\theta_N \theta_H}{\theta_{NH_2}\theta_H} \quad , \quad \frac{p^\circ}{} \quad , \quad \theta_*^2 \, p_{H_2}/p^\circ \Big)$$

Limiting Analysis:

Case 1 : (1) is RLS , (3) is QE

$$K_2 = \frac{\theta_{NH_2}\theta_H}{\theta_{NH}\theta_N} \; ; \; r_{ov} = 2r_1 = 2\left(k_1 P_{N_2} \theta_*^2 - k_{-1}\theta_N^2\right)$$

$$\theta_H = \sqrt{K_5 \frac{P_{H_2}}{p^\circ}}\,\theta_* \; ; \; \theta_{NH_2} = \frac{P_{NH_2}/p^\circ}{K_4\sqrt{K_5 \frac{P_{H_2}}{p^\circ}}}\,\theta_* \qquad P_i' = \frac{P_i}{P_0}$$

$$\theta_{NH} = \frac{\theta_{NH_2}}{K_3\sqrt{K_5 \frac{P_{H_2}}{p^\circ}}} = \frac{P_{NH_3}/p^\circ}{K_3K_4K_5\frac{P_{H_2}}{p^\circ}}\,\theta_* \; ; \; \theta_N = \frac{P'_{NH_3}}{K_2K_3K_4\left(K_5 P'_{H_2}\right)^{3/2}}\,\theta_*$$

$$r = 2k_1 P_{N_2}\theta_*^2\left[1 - \frac{k_{-1}}{k_1}\frac{P_{NH_3}^2}{P_{N_2}K_2^2K_3^2K_4^2K_5^2P_{H_2}^3}\right]$$

$$r_{1(lim)} = \frac{2k_1 P_{N_2}(1-\gamma)}{\left(1+\sqrt{K_5 P'_{H_2}} + \frac{P'_{NH_3}}{K_4\sqrt{K_5 P'_{H_2}}} + \frac{P'_{NH_3}}{K_3K_4K_5P'_{H_2}} + \frac{P'_{NH_3}}{K_2K_3K_4\left(K_5P'_{H_2}\right)^{3/2}}\right)^2}$$

$$\gamma = \frac{1}{k_1K_2^2K_3^2K_4^2K_5^2}\,\frac{P'^2_{NH_3}}{P'_{H_2}P'^3_{N_2}} \; ; \; 0<\gamma<1$$

Case 2 : (3) is RLS , (1) is QE

$$r_{overall} = r_3 = k_3\theta_{NH}\theta_H - k_{-3}\theta_{NH_2}\theta_* \; ; \; K_1 = \frac{\theta_N^2}{\theta_*^2\frac{P_{H_2}}{p^\circ}} \Rightarrow \theta_N = \sqrt{K_1 P_{H_2}}\,\theta_*$$

$$\theta_{NH_3} = \frac{P'_{NH_3}\,\theta_*}{K_4\sqrt{K_5 P'_{H_2}}} \quad ; \quad \theta_N = \sqrt{K_5 P'_{H_2}}\;\theta_*$$

$$\theta_{NH} = K_2\sqrt{K_1 P'_{H_2}}\;\sqrt{K_5 P'_{H_2}}\;\theta_*$$

$$r_{3\,lim} = \frac{k_3 K_2 K_1^{1/2} K_5 \sqrt{P'_{N_2}}\; P'_{H_2}\,(1-\sqrt{\gamma})}{\left(1+\sqrt{K_5 P'_{H_2}}+\sqrt{K_1 P'_{H_2}}+K_2\sqrt{K_1 P'_{N_2}}\sqrt{K_5 P'_{H_2}}+\frac{P'_{NH_3}}{K_4\sqrt{K_5 P'_{H_2}}}\right)^2}$$

Principle: many catalysts follow either $r_{1\,lim}$ , $r_{3\,lim}$ expressions.